

The Double-Weighted Portfolio in the Standard Nuclear oliGARCHy

A Multidimensional Framework for
Asset Allocation and True Risk Premia

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Abstract

This paper embeds the double-weighted portfolio framework into the Standard Nuclear oliGARCHy (SNoG) as a multidimensional architecture for asset allocation, systemic risk, and true risk premia. The double-weighted portfolio is defined by a normalized matrix of allocation weights and a corresponding weighted portfolio value. Previous work establishes the existence of infinitely many admissible allocations, separable and time-dependent constructions, stochastic Dirichlet weights, and stochastic price dynamics. A second paper demonstrates applications to sector–region, asset-class–economic-regime, hierarchical, multi-period, and volatility–correlation portfolios.

Rather than reproducing those results, this paper interprets them within the SNoG state-space. The central object becomes a joint state consisting of the SNoG state and a portfolio-allocation simplex. This permits asset prices, capital allocation, systemic risk, sovereign capability, economic regimes, and strategic states to be represented within a common multidimensional framework. The paper distinguishes conventional estimated risk premia from the proposed SNoG concept of a true premium and outlines an econometric, computational, and decision-theoretic research program. The resulting framework is compatible with portfolio theory, stochastic processes, systemic-risk analysis, dynamic programming, machine learning, and multidimensional sovereign-capability analysis, while leaving empirical identification as a question for future research.

The paper ends with “The End”

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1 Introduction

The central proposition of this paper is simple: the portfolio need not be represented by a single vector of weights over a single cross-section of assets. Allocation may simultaneously occur across several economically meaningful dimensions.

The double-weighted portfolio begins with the normalization condition

$$\sum_{i=1}^m \sum_{j=1}^{n_i} w(i, j) = 1. \quad (1)$$

Its associated portfolio value is

$$P_t = \sum_{i=1}^m \sum_{j=1}^{n_i} w(i, j, t) p(i, j, t). \quad (2)$$

The existence paper establishes that this constraint defines a simplex of admissible allocations and consequently admits infinitely many solutions. It also establishes non-uniqueness when the portfolio price is fixed. These results provide the mathematical foundation for treating allocation itself as a state variable rather than merely as an estimated coefficient vector.

The empirical construction paper demonstrates that the two dimensions can represent economically distinct structures, including sector and region, asset class and economic regime, time, and risk-factor classifications. The hierarchical extension further permits three or more dimensions.

The SNoG problem therefore admits a natural generalization:

$$\mathcal{S}_t = \mathcal{X}_t \times \Delta^{N-1}, \quad (3)$$

where $\mathcal{X}_t$ denotes the SNoG state and Δ^{N-1} denotes the portfolio-allocation simplex.

The purpose of this paper is not to assert that this construction has already been empirically validated. Its purpose is to formulate a coherent mathematical architecture within which such validation can subsequently occur.

2 The Double-Weighted Portfolio

Let the first index represent a primary economic category and the second index represent a subordinate category. The weight matrix is

$$W_t = [w(i, j, t)]. \quad (4)$$

The admissible allocation space is

$$\Delta^{N-1} = \left\{ W_t : w(i, j, t) \geq 0, \sum_{i,j} w(i, j, t) = 1 \right\}, \quad (5)$$

where

$$N = \sum_{i=1}^m n_i. \quad (6)$$

The existence result follows immediately from the geometry of the simplex.

$$\dim(\Delta^{N-1}) = N - 1. \quad (7)$$

Thus, unless additional restrictions are imposed, allocation is inherently non-unique.

This observation is important for SNoG. A unique allocation should not be assumed merely because a unique portfolio price is observed. Identification of the allocation requires additional economic, institutional, statistical, or strategic restrictions.

3 Separable and Stochastic Allocations

A useful special case is the separable construction

$$w(i, j, t) = \alpha_i(t)\beta_j(t), \quad (8)$$

where

$$\sum_i \alpha_i(t) = 1, \quad (9)$$

$$\sum_j \beta_j(t) = 1. \quad (10)$$

Consequently,

$$\sum_i \sum_j w(i, j, t) = 1. \quad (11)$$

This permits an economically useful distinction between two allocation decisions. The first may describe selection among primary assets or capabilities; the second may describe temporal, geographical, regime, strategic, or systemic exposure.

The stochastic construction in the existence paper uses Dirichlet distributed weights:

$$W_t \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_N). \quad (12)$$

By construction,

$$\sum_{k=1}^N w_k(t) = 1 \quad \text{almost surely.} \quad (13)$$

The corresponding stochastic portfolio is

$$P_t = \sum_{i,j} w(i, j, t) p(i, j, t). \quad (14)$$

If prices follow geometric Brownian motion, one possible specification is

$$dp_{ij,t} = \mu_{ij,t} p_{ij,t} dt + \sigma_{ij,t} p_{ij,t} dW_{ij,t}. \quad (15)$$

The resulting construction separates uncertainty in allocation from uncertainty in prices. This is particularly relevant when systemic conditions alter both investment opportunities and the allocation itself.

4 From Portfolio Space to SNoG State Space

The SNoG state is treated here abstractly as a multidimensional state vector

$$X_t = (x_{1,t}, \dots, x_{K,t}). \quad (16)$$

The portfolio allocation is then embedded into the state:

$$Z_t = (X_t, W_t). \quad (17)$$

The resulting state space is

$$\mathcal{Z} = \mathcal{X} \times \Delta^{N-1}. \quad (18)$$

This construction has a direct interpretation.

$$\boxed{\text{SNoG state} + \text{allocation state} = \text{joint financial-system state}} \quad (19)$$

The important change is conceptual. Instead of asking only how an observable price responds to observable risk factors, one can ask how the entire allocation state evolves jointly with the systemic state.

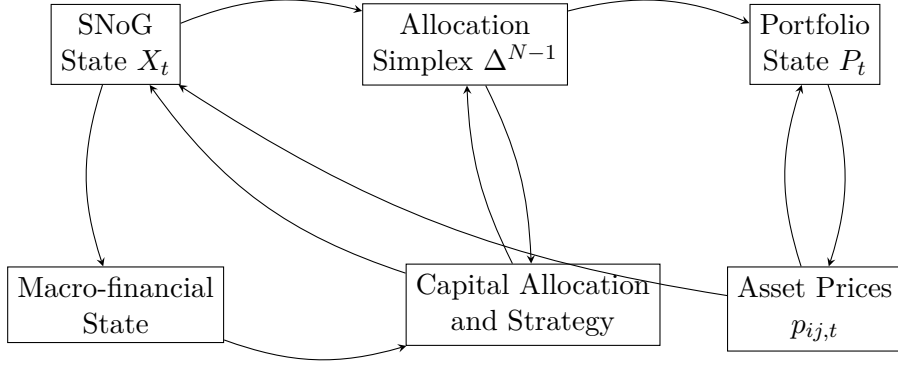


Figure 1: Conceptual architecture of the double-weighted portfolio embedded in the SNoG state space.

5 Multidimensional Economic Interpretation

The empirical portfolio paper provides five useful interpretations of the two-dimensional architecture.

Table 1: Double-weighted portfolio dimensions

First dimension	Second dimension	Economic interpretation
Sector	Region	Geographic sector allocation
Asset class	Economic regime	State-contingent allocation
Asset class	Sector	Hierarchical allocation
Year	Quarter	Multi-period allocation
Volatility	Correlation	Risk-factor diversification

These examples can be generalized for SNoG without requiring that every dimension be financial.

For example, the first dimension may represent sovereign capability and the second financial exposure. Alternatively, the first dimension may represent economic regimes while the second represents strategic assets.

The architecture therefore permits

$$w(i, j, t) = w(\text{economic state}, \text{strategic state}, \text{financial state}, t). \quad (20)$$

A three-dimensional extension is

$$w(i, j, k, t), \quad (21)$$

and a general d -dimensional allocation tensor is

$$w(i_1, \dots, i_d, t), \quad (22)$$

subject to

$$\sum_{i_1} \dots \sum_{i_d} w(i_1, \dots, i_d, t) = 1. \quad (23)$$

This provides a mathematical route from a portfolio matrix to a multidimensional allocation tensor.

6 Asset Pricing and True Risk Premia

Let the stochastic discount factor be M_{t+1} . Standard asset pricing requires

$$1 = E_t [M_{t+1} R_{i,t+1}]. \quad (24)$$

For excess returns,

$$E_t[R_{i,t+1}^e] = -\text{Cov}_t(M_{t+1}, R_{i,t+1}^e). \quad (25)$$

The SNoG extension proposed here is to permit the discount factor and returns to depend on the joint state Z_t :

$$M_{t+1} = M(Z_t, Z_{t+1}, \varepsilon_{t+1}). \quad (26)$$

Likewise,

$$R_{i,t+1} = R_i(Z_t, Z_{t+1}, \varepsilon_{i,t+1}). \quad (27)$$

The term “true premium” is used here for the economically structural premium associated with exposure to the SNoG state, rather than for an arbitrary coefficient produced by a particular empirical regression.

One possible population object is

$$\pi_i^*(Z_t) = E_t[R_{i,t+1}^e | Z_t]. \quad (28)$$

For a finite sample of T observations, a simple estimator is

$$\hat{\pi}_i = \frac{1}{T} \sum_{t=1}^T \hat{\pi}_{i,t}. \quad (29)$$

The use of a mean is therefore an estimation choice, not a theorem that eliminates state dependence.

7 Systemic Risk

Let systemic risk be represented by a scalar functional

$$\mathcal{R}_t = \mathcal{R}(X_t, W_t, \Sigma_t), \quad (30)$$

where Σ_t denotes the conditional covariance structure of asset returns.

The distinction between ordinary portfolio risk and systemic risk is important:

$$\text{portfolio risk} \neq \text{systemic risk}. \quad (31)$$

A portfolio may be individually diversified while remaining exposed to a common systemic state.

One possible systemic-risk decomposition is

$$\mathcal{R}_t = \mathcal{R}_t^{\text{id}} + \mathcal{R}_t^{\text{common}} + \mathcal{R}_t^{\text{SNoG}}. \quad (32)$$

The final component is deliberately left as a research object rather than being assigned an empirical formula in advance.

8 Econometrics: What Regression Can and Cannot Do

Regression remains useful for measurement, estimation, and falsification. It should not, however, be confused with the definition of the economic architecture.

A generic conditional model is

$$R_{t+1}^e = \alpha_t + \beta_t' F_{t+1} + \varepsilon_{t+1}, \quad (33)$$

where the coefficients may themselves depend upon the SNoG state:

$$\beta_t = \beta(X_t, W_t). \quad (34)$$

This produces a state-dependent asset-pricing model rather than a necessarily constant-factor model.

The econometric task is consequently to distinguish among:

$$\text{measurement : estimate observable quantities,} \quad (35)$$

$$\text{identification : determine which structural objects are recoverable,} \quad (36)$$

$$\text{validation : compare model implications with data.} \quad (37)$$

A regression is one instrument within this larger procedure.

9 Recapitalization and Dynamic Allocation

The double-weighted structure naturally accommodates changes in the capital base. Let total capital be C_t . Then capital assigned to cell (i, j) is

$$C_{ij,t} = C_t w(i, j, t). \quad (38)$$

If recapitalization changes the capital stock according to

$$C_{t+1} = C_t + I_t - D_t + G_t, \quad (39)$$

where I_t denotes new investment, D_t distributions or depletion, and G_t other net capital adjustments, then the allocation system becomes dynamically coupled to capital accumulation.

The portfolio value becomes

$$V_t = C_t \sum_{i,j} w(i, j, t) r_{ij,t}. \quad (40)$$

This makes recapitalization a state transition rather than merely an accounting afterthought.

10 Decision Theory and Welfare Economics

A social planner may evaluate the allocation according to

$$\max_{\{W_t\}} E_0 \left[\sum_{t=0}^T \delta^t U(C_t, W_t, X_t) \right]. \quad (41)$$

Subject to

$$w(i, j, t) \geq 0, \quad (42)$$

$$\sum_{i,j} w(i, j, t) = 1, \quad (43)$$

$$C_{t+1} = \mathcal{C}(C_t, W_t, X_t, \varepsilon_{t+1}). \quad (44)$$

This formulation allows portfolio decisions to be evaluated not only by private expected return but also by welfare, resilience, distributional effects, and systemic externalities.

The framework therefore connects naturally to welfare economics and public-finance questions without requiring that private financial returns be equated with social welfare.

11 Political Economy and Geopolitical States

The SNoG extension introduces a domain in which financial allocation and sovereign strategic states may interact. The paper does not assume that political power can be reduced to a financial variable.

Instead, define a capability vector

$$K_t = (K_t^E, K_t^F, K_t^M, K_t^D, K_t^T, K_t^I), \quad (45)$$

representing, for example, economic, financial, military, diplomatic, technological, and institutional capabilities.

The joint state may then be written as

$$Z_t = (X_t, W_t, K_t). \quad (46)$$

International relations, diplomacy, and strategic interaction enter through the transition law

$$Z_{t+1} = \Phi(Z_t, a_t, \eta_{t+1}), \quad (47)$$

where a_t denotes strategic or policy actions.

This is a state-transition representation, not a claim that financial markets mechanically determine geopolitical outcomes.

12 Warfare Economics and Strategic Allocation

In a conflict-sensitive environment, the allocation constraint remains

$$\sum_{i,j} w(i, j, t) = 1, \quad (48)$$

but the objective may change.

A stylized development allocation is

$$U_t^D = U(C_t^{\text{civil}}, C_t^{\text{productive}}, X_t), \quad (49)$$

while a strategic-resilience allocation may be represented by

$$U_t^S = U(C_t^{\text{security}}, C_t^{\text{resilience}}, X_t). \quad (50)$$

A switching formulation can therefore be written as

$$U_t = (1 - s_t)U_t^D + s_t U_t^S, \quad s_t \in [0, 1]. \quad (51)$$

The framework does not imply that warfare is desirable. It provides a mathematical representation for studying how resource allocation changes when strategic conditions change.

13 Physics, Biology, Psychology, and Complex Systems

The double-weighted architecture is mathematically compatible with multiscale systems in other disciplines.

In physics, the state-space representation resembles a coupled dynamical system:

$$\dot{Z}_t = F(Z_t, t). \quad (52)$$

In biology, allocation may represent competing flows of resources among organs, populations, or ecological niches:

$$\sum_i x_i(t) = C(t). \quad (53)$$

In psychology and behavioural economics, the weights may represent attention, preference, salience, or perceived risk:

$$w_i(t) = \frac{\exp(\theta_i(t))}{\sum_j \exp(\theta_j(t))}. \quad (54)$$

These analogies should not be mistaken for empirical equivalence. Their value is architectural: the same normalized allocation mathematics can represent resource competition in heterogeneous systems.

14 Computer Science and Algorithms

The framework is naturally representable as a stateful computational system.

Algorithm: Double-Weighted SNoG Allocation

Input:

SNoG state X_t
Asset-price matrix P_t
Capital C_t
Allocation parameters A_t

1. Construct admissible weight matrix W_t .
2. Normalize W_t so that $\text{sum}(W_t) = 1$.
3. Observe or simulate P_t .
4. Compute portfolio value V_t .
5. Update systemic state X_{t+1} .
6. Update capital C_{t+1} .
7. Reallocate W_{t+1} .
8. Estimate state-dependent risk and premium objects.
9. Store $(X_t, W_t, P_t, V_t, C_t)$ for validation.
10. Repeat.

The same structure can support dynamic programming:

$$V(Z_t) = \max_{a_t} \{U(Z_t, a_t) + \delta E_t[V(Z_{t+1})]\}. \quad (55)$$

It can also be implemented with machine-learning models, provided that machine learning is used as an estimation or approximation layer rather than silently replacing the underlying economic definitions.

15 AI and Machine Learning

A learned transition function may take the form

$$\hat{Z}_{t+1} = f_{\theta}(Z_t, a_t, \varepsilon_t). \quad (56)$$

A neural approximation to the allocation policy can be written as

$$W_t = \text{softmax}(f_{\theta}(Z_t)). \quad (57)$$

The softmax transformation automatically enforces positivity and normalization. An alternative is to learn the conditional premium function:

$$\hat{\pi}_i = g_{\theta}(X_t, W_t, K_t). \quad (58)$$

The resulting architecture can combine statistical estimation, state-space modeling, dynamic programming, and machine learning without requiring any one technique to be treated as universally sufficient.

16 Identification and Falsifiability

The existence paper establishes mathematical existence and non-uniqueness; these results do not constitute empirical validation.

The distinction can be summarized as follows.

Table 2: Levels of evidence in the framework

Level	Question	Evidence
Mathematical	Does a solution exist?	Proof
Structural	Is the model well-defined?	Definitions and constraints
Statistical	Does the model fit data?	Estimation and diagnostics
Predictive	Does it forecast?	Out-of-sample testing
Causal	Does an intervention matter?	Identification strategy
Economic	Does it improve decisions?	Welfare or utility analysis
Institutional	Does it describe institutions?	Historical and institutional evidence

Consequently, the appropriate empirical program is not simply to run a regression. It is to establish whether the proposed state variables can be measured, whether their transitions can be identified, and whether their asset-pricing implications survive out-of-sample tests.

17 Proof Sketches

The principal mathematical claims needed here follow directly from the source framework.

17.1 Existence

For any positive integers m, n , the uniform allocation

$$w(i, j) = \frac{1}{mn} \quad (59)$$

satisfies

$$\sum_{i=1}^m \sum_{j=1}^n w(i, j) = 1. \quad (60)$$

Therefore the admissible allocation space is non-empty.

17.2 Non-Uniqueness

The feasible set is an $(mn - 1)$ -dimensional simplex. For $mn > 1$, it contains infinitely many points. Therefore normalization alone cannot identify a unique allocation.

17.3 Separable Solutions

If

$$\sum_i \alpha_i = 1, \quad (61)$$

$$\sum_j \beta_j = 1, \quad (62)$$

then

$$\sum_i \sum_j \alpha_i \beta_j = \left(\sum_i \alpha_i \right) \left(\sum_j \beta_j \right) = 1. \quad (63)$$

Hence $w(i, j) = \alpha_i \beta_j$ is admissible.

18 Research Program

A practical empirical program can proceed through the following stages.

1. Define the SNoG state variables.
2. Construct the relevant double-weighted allocation matrices.
3. Determine which variables are observable and which are latent.
4. Estimate the allocation process.
5. Estimate conditional price and return dynamics.
6. Construct systemic-risk measures.
7. Estimate state-dependent true-premium objects.
8. Compare the structural model against conventional asset-pricing benchmarks.
9. Perform out-of-sample validation.
10. Conduct sensitivity, robustness, and counterfactual analysis.

The central computational object is therefore not a single regression but a joint system:

$$\boxed{(X_t, W_t, C_t, P_t) \longrightarrow (X_{t+1}, W_{t+1}, C_{t+1}, P_{t+1})} \quad (64)$$

with empirical observations providing partial information about the latent state.

19 Conclusion

The double-weighted portfolio provides a mathematically simple but structurally rich extension of conventional portfolio representation. Its central constraint generates a simplex of admissible allocations, while its matrix structure permits economically meaningful dimensions to be represented simultaneously.

The principal SNoG extension is to treat the allocation matrix as part of the state itself:

$$\mathcal{S}_t = \mathcal{X}_t \times \Delta^{N-1}. \quad (65)$$

This produces a multidimensional architecture in which systemic state, capital allocation, asset prices, economic regimes, strategic conditions, and risk premia may interact.

The framework does not establish that SNoG generates empirically superior asset-pricing predictions. That question remains open. What it does establish, building on the double-weighted portfolio results, is a well-defined mathematical environment in which such a question can be posed.

The methodological lesson is therefore:

Architecture $\rightarrow$ State $\rightarrow$ Allocation $\rightarrow$ Prices $\rightarrow$ Risk $\rightarrow$ Premium

 (66)

Regression, stochastic calculus, optimization, simulation, econometrics, machine learning, and computational methods can subsequently operate within that architecture. They need not be treated as competing worldviews.

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Glossary

Double-weighted portfolio A portfolio whose allocation is indexed by two dimensions and whose weights satisfy a normalization constraint.

Allocation simplex The set of non-negative allocation vectors or matrices whose elements sum to one.

SNoG Standard Nuclear oliGARCHy, the systemic state-space framework within which the present portfolio architecture is embedded.

SNoG state The multidimensional state describing the relevant systemic, economic, financial, and strategic conditions.

True premium The structural premium associated with exposure to the relevant economic state, distinguished from a coefficient that merely estimates an empirical regression relationship.

Systemic risk Risk arising from common or interconnected states whose effects extend beyond the idiosyncratic risk of an individual asset.

Recapitalization An increase or restructuring of the capital available to an economic or financial system.

State space The set of possible states of a dynamic system together with its transition structure.

Dirichlet distribution A probability distribution over non-negative vectors whose components sum to one, making it useful for stochastic portfolio weights.

Geometric Brownian motion A continuous-time stochastic process frequently used as a stylized model of asset-price dynamics.

State-dependent premium A risk premium whose magnitude depends upon the prevailing economic or systemic state.

Allocation tensor A multidimensional generalization of the allocation matrix.

The End